LEO Station - Structural Engineering Calculations

for the 3D Model

1. Solar Panel System Design

1.1 Power Requirements Analysis (NASA-Referenced)

Life Support Power Requirements:

- Source: NASA OIG. (2021). NASA's Management of the International Space Station
- ISS Life Support (6 crew): 15 kW baseline
- Scaling for 10 persons: 15 x (10/6) = 25 kW
    - O_2 generation: 8 kW
    - CO_2 removal: 5 kW
    - Water recycling: 7 kW
    - Air circulation: 5 kW

Agriculture System Power Requirements:

- Source: Jones, H. (2018). Hydroponics for Food Production in Space NASA
- LED Requirements: 1000 μmol m^{-2} s^{-1} (NASA standard)
- Agriculture Module: 30 kW
    - LED grow lights: 25 kW
    - Nutrient pumps: 2 kW
    - Climate control: 3 kW

Station Operations:

- Source: NASA Thermal Control Guidlines
- Baseline Systems: 12 kW
    - Communications: 3 kW
    - Computers/control: 4 kW
    - Lightning/misc: 5 kW

Tourist Facilities:

- Conservative Estimate: 6 kW
    - Entertainment systems: 2 kW
    - Exercise equipment: 2 kW
    - Cameras/recording: 2 kW

Total Peak Power: 73 kW
Continuous Power (NASA safety margin): 88 kW

1.2 Solar Panel Specifications (NASA Standards)

Source: NASA.(1995). Solar Power System Environmental Durability
Solar Panel Efficiency in LEO:

- Space-grade panels: 30% efficiency (NASA specification)
- Solar constant: 1,361 W/m^2 (NASA standard)
- Orbital efficiency factor: 0.6 (accounts for eclipse periods)
- Effective solar irradiance: 816 W/m^2

Required Panel Area:
- Power per m^2: 816 x 0.30 = 245 W/m^2
- Required area: 88,000 W ÷ 245 W/m^2 = 359 m^2

Panel Configuration:
- Standard space panel: 2.5m x 1.2m = 3m^2 each
- Number of panels needed: 359 ÷ 3 = 120 panels
- Arranged in 4 deployable arrays of 30 panels each

1.3 Deployment Constraints (Falcon 9 Requirements)

Source: SpaceX Falcon 9 Users Guide (2021)
- Fairing diameter: 5.2m maximum
- Stowed dimensions: Must fit within fairing
- Deployment mechanism: Accordion-style fold
- Final deployed span: 30 m x 12m per array

2. Pressure Vessel Design (NASA Safety Standards)

2.1 Design Parameters

Source: AFSPCMAN 91-710 Range Users Manual & NASA safety
Requirements Operating Conditions:
- Internal pressure: 101.3 kPa (1 atmosphere)
- External pressure: 0 kPa (vacuum)
- NASA Safety factor: 4.0 (mandatory for human-rated systems)
- Design pressure: 405.2 kPa

Material Properties (NASA-approved Aluminium 2219-T87):
- Ultimate tensile strength: 455 MPa
- Yield strength: 365 MPa
- Density: 2,840 kg/m^3
- Design stress: 91.25 MPa (yield strength ÷ 4)

2.2 Cylindrical Section Calculation

Formula: $t = (P \times r) / (\sigma \times \eta)$ Where:

- P = design pressure = 405.2 kPa
- r = radius = 2.5m (5.0m diamater)
- σ = allowable stress = 91.25 MPa
- η = joint efficiency = 0.85

Calculation: $t = (405.2 \times 2.5) / (91{,}250 \times 0.85) = 0.0131$ m = 13.1 mm
Design thickness: 15 mm (rounded up for manufacturing)

2.3 Spherical End Caps

Formula: $t = (P \times r) / (2 \times \sigma \times \eta)$
Calculation: $t = (405.2 \times 2.5) / (2 \times 91{,}250 \times 0.85) = 6.5$ mm
Design thickness: 10 mm (minimum for structural stability)

3. Docking Mechanism System (NASA CBM Standards)

3.1 Common Berthing Mechanism (CBM) Specifications

Source: NASA ISS Interface Control Documents
NASA Standard CBM Requirements:

- Maximum docking load: 890 kN (200,000 lbf)
- Lateral load capability: 445 kN (100,000 lbf)
- Operating temperatur: -157°C to +121°C
- Proven on ISS since 2000

3.2 Module Interface Design (NASA-Compliant)

Connection points per Module:

- Primary CBM: 1 (for main station connection)
- Secondary CBM: 1-2 (for expansion modules)
- Total Connections: 8 CBM for complete station

Structural Requirements (NASA Specification):

- Bolt circle diameter: 1.27m
- Number of bolts: 16 x M20 grade 8.8
- Preload per bolt: 180 kN
- Total preload capacity: 2,880 kN

4. MMOD Debris Protection (NASA Standards)

4.1 NASA MMOD Protection Standards

Source: NASA Orbital Debris Mitigation Standard Practices (ODMSP)

Target Protection Level:

- Practical size: up to 1 cm diameter
- Velocity: up to 15 km/s
- Material: Aluminium sphere equivalent
- Collision probability must be <0.001 over 100 years

4.2 Whipple Shield Configuration (NASA-Proven Design)

Source: NASA Hypervelocity Impact Technology Facility (HITF)
Layer 1- Outer Bumber:

- Material: Aluminium 2024-T3 (NASA standard)
- Thickness: 3 mm
- Spacing from pressure wall: 150 mm

Layer 2 - Stuffing:

- Material: Nextel/Kevlar fabric (NASA deployed)
- Thickness: 25 mm
- Areal density: 6kg/m^2

Layer 3 - Pressure Wall:

- Material: Aluminium 2219-T87
- Thickness: 15 mm (from pressure calculation)

4.3 Coverage Area (based on NASA Risk Assessment)

Critical Areas:

- Inhabited modules: 100% coverage (NASA requirement)
- Agriculture modules: 90% coverage
- Service modules: 75% coverage
- Total shielded area: 800m^2

Mass impact:

- Outer bumper: 800 x 3 x 0.003 x 2,700 = 19.4 tons
- Stuffing: 800 x 6 = 4.8 tons
- Total shielding mass: 24.2 tons

5. Falcon 9 Integration & Mass Analysis

5.1 Fairing Constraints (SpaceX User Guide)

Source: SpaceX Falcon 9 Users Guide Section 12
Standard Fairing Dimensions:

- Outer diameter: 5.2 m (17.2 ft)
- Height: 13.2 m (43.5 ft)
- Payload static envelope: See Figure 12-5
- Access doors: 1 standard (610 mm diameter)

Payload Interface Options:
- Standard PAF: 1,575 mm (62.01”) bolted interface
- Large PAF: 2,624 mm (103.307”) bolted interface
- Alternative interfaces: 937 mm, 1,194 mm, 1,666 mm available

5.2 Module Configuration Analysis

Module Type A - Habitat Modules (Tourist & Crew Quarters):
- Dimensions: 5.0 m diameter x 8 m length
- Fits in 5.2 m x 13.2 m fairing
- Primary structure mass: 10 tons
- Life support systems: 2 tons
- Interior/furniture: 1.5 tons
- Total Module Mass: 16.5 tons

Module Type B - Agriculture Module:
- Dimensions: 5.0 m diameter x 6 m length
- Fits in 5.2 m x 13.2 m fairing
- Primary structure mass: 8 tons
- MMOD shielding: 2.5 tons
- Aeroponic systems: 3 tons
- LED arrays: 1.5 tons
- Total Module Mass: 15 tons

Module Type C - Service Module (Power/Thermal):
- Dimensions: 5.0 m diameter x 6 m length
- Fits in 5.2 m x 13.2 m fairing
- Primary structure mass: 8 tons
- MMOD shielding: 2 tons
- Power systems: 3 tons
- Thermal control: 2 tons
- Total Module Mass: 15 tons

Solar Panel Arrays (folded):
- Stowed dimensions: 4.5 m x 2 m x 1 m
- Fits in 5.2 m x 13.2 m fairing
- Array mass: 1.5 tons each
- 4 arrays total: 6 tons total

5.3 Center of Gravity Analysis (NASA/SpaceX Requirements)

Module Type A - Habitat (16.5 tons):
- CoG location: 4.0 m from base
- Within Falcon 9 PAF capability (Figure 3-3)

Module Type B - Agriculture (15 tons):
- CoG location: 3.0m from base

- Within Falcon 9 PAF capability

Module Type C - Service (15 tons)
- CoG location: 3.0 m from base
- Within Falcon 9 PAF capability

Solar Arrays (1.5 tons each):
- CoG location: 1.0 m from base
- Well within limits

5.4 Launch Sequence & Assembly Order

Launch Schedule (8 Falcon 9 missions):

Mission 1: Core Service Module + 2 Solar Arrays
- Combined mass: 18 tons
- First module establishes power/thermal baseline

Mission 2: Primary Habitat Module
- Mass: 16.5 tons
- Docks to Service Module via CBM

Mission 3: Agriculture Module
- Mass: 15 tons
- Docks to create T-configuration

Mission 4: Secondary Habitat Module
- Mass: 16.5 tons
- Extends habitat capacity

Mission 5: Remaining 2 Solar Arrays
- Mass: 3 tons
- Complete power system

Mission 6-7: Expansion modules as demand grows
- additional tourism modules, research labs

5.5 Structural Load Verification

Launch Load Factors (SpaceX Falcon 9 Users Guide):
- Axial: +6g /-3g
- Lateral: ±2g
- All modules designed for these loads

Frequency Requirements:
- Primary lateral: > 10Hz
- Primary axial: >25Hz

- Secondary structure: >35Hz

6. Final Engineering Specifications for the 3D Model

6.1 Core Station Configuration

- **Total assembled mass**: 84 tons
- **Total volume:** 1,600 m^3
- **Power generations:** 88 kW
- **Crew capacity:** 10 persons
- **Module Count:** 4 core + 2-3 expansions

6.2 Critical Dimensions for the 3D Model

- **Module diameter:** 5.0 m (maximum)
- **Module lengths:** 6 m, 8 m variants
- **CBM docking ports:** 1.27 m diameter
- **Solar arrays:** 30 m x 12 m deployed
- **Wall thickness:** 15 mm + 3 mm MMOD shield

6.3 Assembly Joints

- 6 x CBM connections (NASA standard)
- Redundant power/data pathways
- Emergency isolation capability

6.4 Launch Compatible Summary

- 6-7 Falcon 9 launches required
- Total launch cost: $375-$435M
- Assembly time: 12-15 months
- All modules fit in standard fairing

7. RESOURCESSSSSSSSSSSS

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